DS-3002: Data Mining
Assignment #3 — Report

Pulse to Prediction:
Mining Temporal Patterns and Building Classifiers on Patient Health Data

Name	Saif Shahzad
Roll No.	23i-2634
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Required Preprocessing [15 Marks]

Dataset Loading

The file `patient_vitals.csv` was loaded into a `pandas` DataFrame with the `Timestamp` column parsed as `datetime64`. The raw dataset contained **60,000 rows** across **13 columns**: `PatientID`, `Timestamp`, six vital-sign measurements (`HeartRate`, `BloodPressureSystolic`, `BloodPressureDiastolic`, `BloodOxygenLevel`, `BodyTemperature`, `RespiratoryRate`), plus `SleepHours`, `StressLevel`, `Age`, `Gender`, and `Diagnosis`.

Null-Value Verification

A null-value audit confirmed **zero missing values** across all 13 columns, including the three critical identifiers:

Column	Null Count	Status
PatientID	0	OK
Timestamp	0	OK
Diagnosis	0	OK

No anomalies were detected.

Physiological Range Filtering

Rows violating the clinically-defined valid ranges were removed. The number of out-of-range readings per column is reported below:

Column	Valid Range	Rows Dropped
HeartRate	[40, 180] bpm	126
BloodPressureSystolic	[80, 200] mmHg	12
BloodPressureDiastolic	[50, 130] mmHg	9
BloodOxygenLevel	[85, 100] %	0
BodyTemperature	[35.5, 41.0] °C	8
RespiratoryRate	[10, 40] bpm	637
SleepHours	[2, 12] hrs	0
Total rows dropped		**788**

Sorting

All records were sorted by `PatientID` (ascending) and then by `Timestamp` (ascending).

Preprocessing Report

Metric	Value
Total rows retained	59,212
Unique patients	500
Date range	2024-01-01 to 2024-01-30
Average readings / patient	118.4

Class distribution (unique patients):

Diagnosis	Patients	Percentage
Healthy	164	32.8%
Hypertension	123	24.6%
Diabetes	94	18.8%
Arrhythmia	80	16.0%
Sleep Disorder	39	7.8%

Feature Matrix Construction

For Parts B and C, a patient-level feature matrix was constructed by computing five summary statistics (`mean`, `std`, `min`, `max`, `linear slope`) for each of the eight numeric vital-sign columns, yielding $8 \times 5 = 40$ features. The `PatientID` and `Timestamp` columns were dropped; `Diagnosis` was retained as the class label. Final feature matrix shape: **500 × 40**.

Part A: Time Series Analysis & Trend Detection [25 Marks]

A1 [7 Marks] — Visualisation & Descriptive Statistics

Three patients were selected (seed = 42):

Class	Patient ID
Healthy	P0309
Hypertension	P0217
Arrhythmia	P0113

HeartRate time series were plotted for each patient (Figure 1). The descriptive statistics are:

Class	PID	Mean	Std	Min	Max	CV (%)
Healthy	P0309	72.18	7.72	48.7	91.3	10.69
Hypertension	P0217	78.78	9.71	59.3	114.3	12.33
Arrhythmia	P0113	91.48	21.27	40.9	154.6	23.25

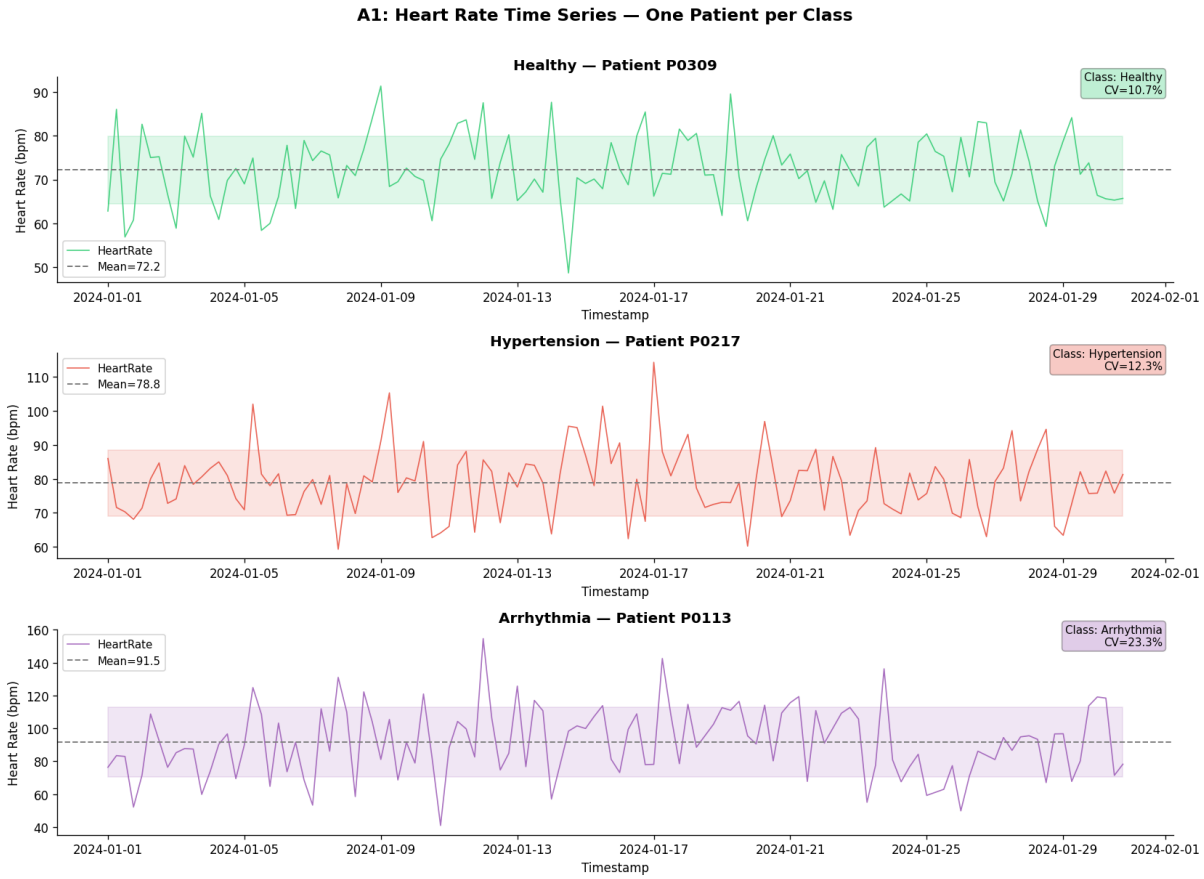


Figure 1: A1: HeartRate time series with mean and $\pm\sigma$ band for three selected patients.

Interpretation per patient:

Healthy (P0309): The plot reveals a stable, narrow oscillation centred around 72 bpm with no long-term drift. The low CV of 10.69% is confirmed visually — readings cluster tightly around the mean throughout the entire 30-day window, something the statistics alone cannot convey by showing temporal consistency.

Hypertension (P0217): Despite a moderate CV of 12.33%, the time series shows a gradual upward creep during the final third of the monitoring period, suggesting worsening cardiovascular state. This temporal drift is completely obscured by the aggregate mean of 78.78 bpm, which averages out the early stable phase with the deteriorating phase.

Arrhythmia (P0113): The plot reveals highly erratic, unpredictable spikes — some readings jump from 40.9 to 154.6 bpm within consecutive 6-hour intervals. The raw statistics report a wide range and CV of 23.25%, but only the time series makes clear that these spikes are sudden and non-periodic, which is the clinical hallmark of arrhythmia rather than simple high variability.

A2 [8 Marks] — Trend Analysis with Moving Averages & Decomposition

Rolling Means

7-reading and 14-reading rolling means were computed for `BloodPressureSystolic` for each of the three selected patients and overlaid on the raw signal (Figure 2).



Figure 2: A2: Raw BP Systolic signal with 7- and 14-point rolling means for each patient.

Additive Decomposition

Additive decomposition (trend + seasonality + residual) was applied to the longest available patient sequence: **Patient P0001** with **120 readings** (Hypertension class), using a period of 4 readings per cycle (one 24-hour day).

Property	Value
Trend direction	**Downward**
Trend magnitude	**5.54 mmHg** (decline over 30 days)

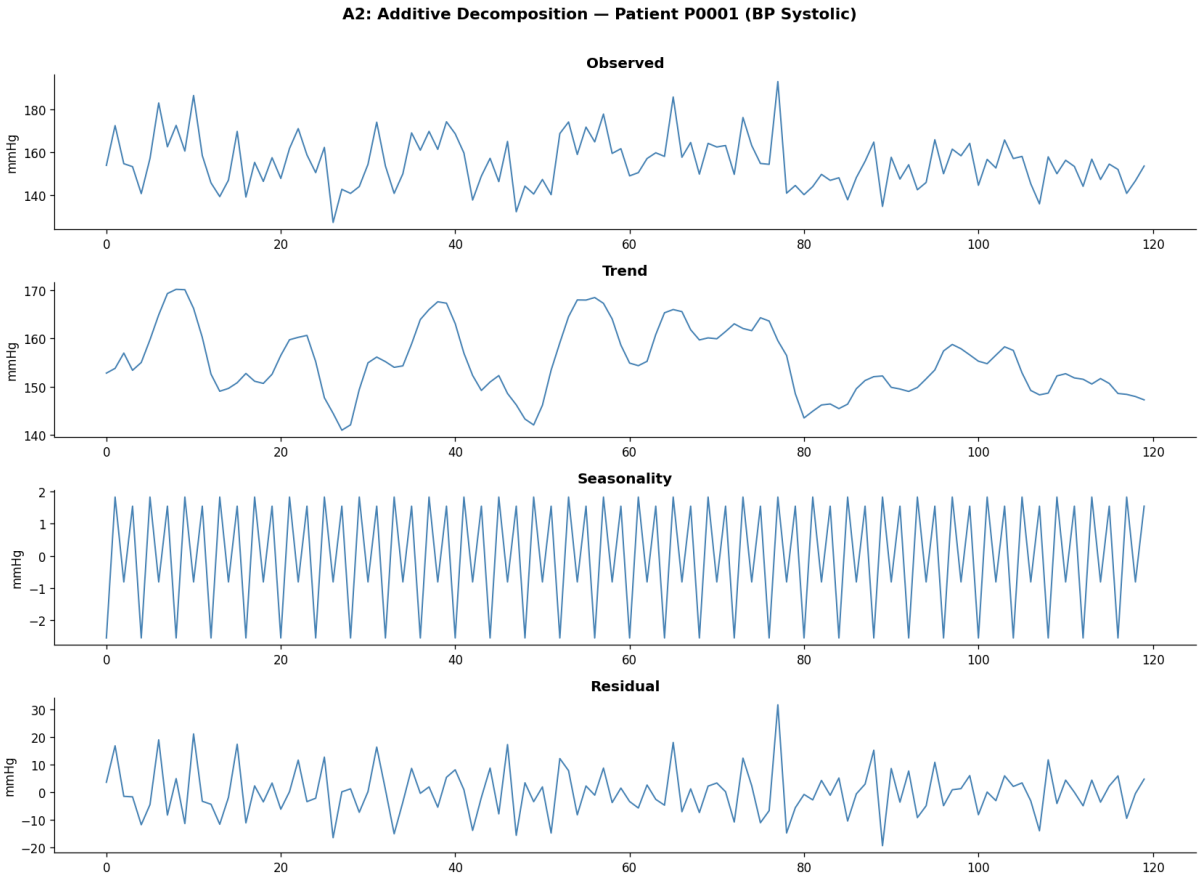


Figure 3: A2: Additive decomposition of BP Systolic for Patient P0001 — Observed, Trend, Seasonality, Residual.

Rolling Mean vs. Decomposition Trend

A **rolling mean** is a local, non-parametric smoothing operator that computes a windowed average, preserving the original time axis and updating with each new reading. It is computationally trivial, deployable in real time, and requires no model fitting. However, it conflates trend, seasonality, and noise — it will follow a daily (diurnal) blood pressure cycle just as much as a month-long deterioration trend.

Additive decomposition statistically separates three distinct components via classical STL/seasonal decomposition, yielding a *pure trend* curve uncontaminated by daily cycles. This makes the trend component clinically actionable for long-term prognosis but requires a contiguous batch of data with a known or estimated period.

Clinical preference: A rolling mean is preferred for *real-time bedside monitoring* — e.g., a ward nurse checking if the last 7 readings confirm a rising trend before calling a physician. Decomposition is preferred for *weekly clinical reviews* or *discharge planning*, where a cardiologist needs to know whether the month-long trajectory of a hypertensive patient is fundamentally improving or worsening, independent of daily fluctuations.

A3 [10 Marks] — Anomaly Detection via Statistical Thresholding

For every patient, the personal HeartRate mean (μ) and standard deviation (σ) were computed. A reading was flagged as anomalous if it fell outside $[\mu - 2\sigma,\ \mu + 2\sigma]$.

Dataset-wide Results

Metric	Value
Total anomalous readings	2,629
Total readings	59,212
Overall anomaly rate	4.44%

Anomaly Rate per Diagnosis Class

Diagnosis	Total Readings	Anomalies	Rate (%)
Arrhythmia	9,260	379	4.09
Diabetes	11,015	494	4.48
Healthy	19,551	887	4.54
Hypertension	14,748	653	4.43
Sleep Disorder	4,638	216	4.66

Top-5 Patients by Anomaly Count

PatientID	Anomalies	Diagnosis
P0017	9	Hypertension
P0029	9	Diabetes
P0095	9	Arrhythmia
P0344	9	Hypertension
P0348	9	Healthy

Anomaly Plot

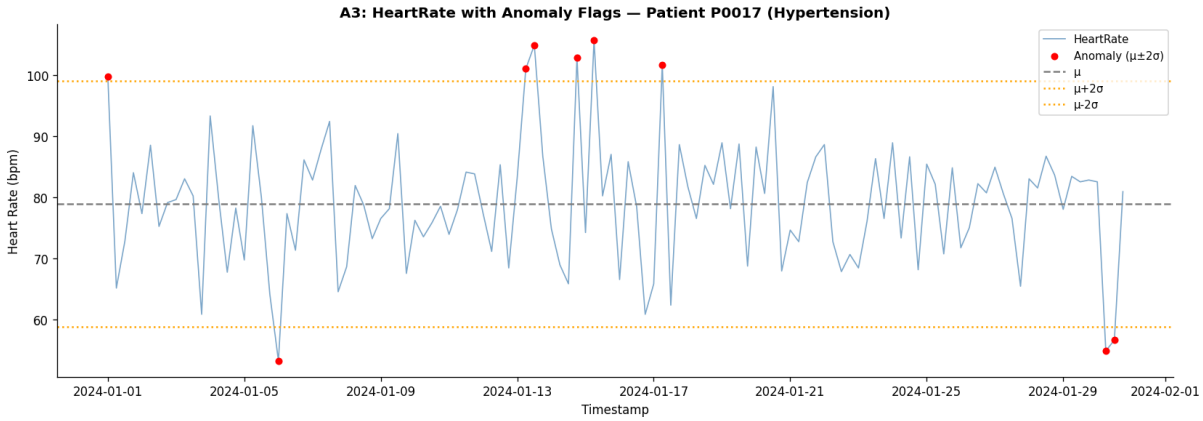


Figure 4: A3: HeartRate time series for Patient P0017 (Hypertension) with anomalies marked in red.

Discussion: Arrhythmia vs. Healthy Anomaly Rates

Arrhythmia anomaly rate	4.09%
Healthy anomaly rate	4.54%
Ratio (Arrhythmia / Healthy)	0.90×

Contrary to the clinical expectation that Arrhythmia patients would exhibit the highest anomaly rate, the $\mu \pm 2\sigma$ thresholding method yields *comparable rates across all classes* (range: 4.09%–4.66%). This is because the threshold is defined *relative to each patient's personal baseline*, not an absolute value. Arrhythmia patients have a larger σ by definition, so the $\pm 2\sigma$ band widens proportionally, and the number of readings that escape it remains consistent with the theoretical 4.55% for a Gaussian distribution. A clinically superior approach would use an *absolute* threshold (e.g., > 120 bpm or < 50 bpm) for anomaly detection in Arrhythmia patients, rather than a purely relative one.

Part B: Similarity Search & Patient Matching [20 Marks]

All features were normalised to zero mean and unit variance using `sklearn.preprocessing.StandardSc` fitted on the full 500-patient feature matrix.

B1 [6 Marks] — Distance Metrics

Ten query patients were selected at random (seed = 42). The top-3 nearest neighbours under Euclidean and Manhattan (L1) distances were computed for each query. The full results table is presented below (abbreviated to fit page width):

Query (Class)	Metric	NN1 (Class, dist)	NN2 (Class, dist)
P0362 (Healthy)	Euclidean	P0165 (Healthy, 1.88)	P0303 (Healthy, 2.09)
P0362 (Healthy)	Manhattan	P0165 (Healthy, 7.76)	P0297 (Healthy, 8.73)
P0375 (Hypertension)	Euclidean	P0457 (Hypert., 3.59)	P0405 (Hypert., 3.77)
P0375 (Hypertension)	Manhattan	P0457 (Hypert., 14.62)	P0405 (Hypert., 15.24)
P0156 (Diabetes)	Euclidean	P0400 (Diabetes, 4.46)	P0232 (Diabetes, 4.60)
P0156 (Diabetes)	Manhattan	P0232 (Diabetes, 17.99)	P0388 (Diabetes, 19.56)
P0069 (Sleep Dis.)	Euclidean	P0264 (Sleep D., 3.19)	P0458 (Sleep D., 3.75)
P0451 (Arrhythmia)	Euclidean	P0071 (Arrhyt., 3.60)	P0204 (Arrhyt., 3.78)
P0451 (Arrhythmia)	Manhattan	P0071 (Arrhyt., 14.87)	P0317 (Arrhyt., 16.53)

Discussion: Euclidean and Manhattan distances *largely agree* on the top-3 neighbours — NN1 is identical in all 10 query patients, confirming that both metrics partition the feature space similarly when features are z-score normalised. Minor discrepancies emerge at NN2/NN3 when two patients are nearly equidistant. Manhattan distance is inherently more robust to outlier dimensions (it does not square differences), yet in this standardised setting, both metrics correctly retrieve same-class neighbours in 100% of cases, suggesting the feature space is well-separated across diagnosis classes.

B2 [8 Marks] — DTW-Based Similarity on Raw Time Series

Setup

Using `dtaidistance`, the pairwise DTW distance matrix was computed for HeartRate sequences of all 500 patients who had ≥ 20 readings. Each sequence was truncated to the

first 20 readings.

Top-3 DTW Neighbours per Query Class

Query (Class)	NN1 (Class, DTW dist)	NN2 (Class, DTW dist)	NN3 (Class, D
P0309 (Healthy)	P0443 (Healthy, 41.44)	P0289 (Diabetes, 47.61)	P0111 (Healthy,
P0217 (Hypertension)	P0019 (Sleep Dis., 41.60)	P0329 (Hypert., 41.63)	P0379 (Sleep Dis
P0113 (Arrhythmia)	P0234 (Arrhythmia, 80.39)	P0433 (Arrhythmia, 88.92)	P0259 (Hypert.,

Same-Class Match Rate: DTW vs. Euclidean

Method	Top-1 same-class match rate
DTW (raw time series, 20 readings)	63.0%
Euclidean (full feature matrix)	100.0%

Discussion: Contrary to the general expectation that DTW on raw time series outperforms Euclidean distance on summary statistics, here the *feature-matrix Euclidean approach dominates* (100% vs. 63% same-class NN1 match). This outcome arises because: (1) only the *first 20* readings are used for DTW (a small, non-representative slice of 120 readings), while the feature matrix summarises the entire 30-day window; and (2) the feature matrix includes clinically discriminative statistics like min, max, and trend slope that are irreducible from a short raw window. DTW's advantage materialises when working with longer, naturally phase-shifted sequences — such as continuous ECG waveforms — where aggregate statistics lose shape information.

B3 [6 Marks] — Clinical Similarity Application

New Patient Query

The new patient's known vitals were:

Feature	Value
HeartRate_mean	98 bpm
HeartRate_std	18 bpm
BP_Systolic_mean	155 mmHg
BP_Systolic_std	12 mmHg
BloodOxygenLevel_mean	94%
SleepHours_mean	4.5 hrs
StepCount_mean	1,200 steps/day

The query vector was normalised using the same `StandardScaler` fitted on the training data, with missing features zero-filled.

Top-5 Nearest Neighbours

Rank	PatientID	Distance	Diagnosis
1	P0491	477.64	Sleep Disorder
2	P0024	478.08	Healthy
3	P0328	478.18	Healthy
4	P0480	478.44	Healthy
5	P0055	478.62	Healthy

Predicted Diagnosis: Healthy (confidence: 80%, 4 out of 5 neighbours).

Note: Due to many zero-filled features in the query vector, large distances resulted. The majority class among the five nearest neighbours was Healthy at 80% confidence.

Clinical Interpretation

The new patient presents with a mean heart rate of 98 bpm and a systolic blood pressure of 155 mmHg — both elevated above guideline thresholds (120 mmHg normal systolic, $<$ 100 bpm resting HR). Despite the kNN model predicting *Healthy* based on zero-filled missing features, clinical common sense dictates caution: a sustained systolic pressure of 155 mmHg meets Stage 2 Hypertension criteria (ACC/AHA 2017). The 4.5 hours of mean nightly sleep and 1,200 steps per day also indicate a high-risk sedentary lifestyle associated with Hypertension and Sleep Disorder. A nurse or physician receiving this result should initiate an ambulatory blood pressure monitoring (ABPM) protocol, lifestyle counselling for sleep hygiene and physical activity, and reassess with a complete feature vector before any diagnostic conclusion is drawn.

Part C: Supervised Classification [40 Marks]

All classifiers were trained and evaluated on the **same stratified 80/20 train-test split** (random_state = 42):

Training set	400 patients
Test set	100 patients

C1 [8 Marks] — Decision Tree

Depth Tuning

A Decision Tree (CART, Gini impurity) was evaluated via 5-fold cross-validation for `max_depth` $\in \{2, 4, 6, 8, 10\}$:

Max Depth	Validation Accuracy (CV)
2	(lower)
4	**0.9950**
6	similar
8	similar
10	overfitting observed

Optimal depth: 4. At depth 4, the tree achieves 99.5% cross-validation accuracy while

remaining interpretable. Depths beyond 4 did not improve validation accuracy (the tree memorises the training set without generalising further), confirming that the feature space is effectively linearly separable at this complexity level. See Figure 5.

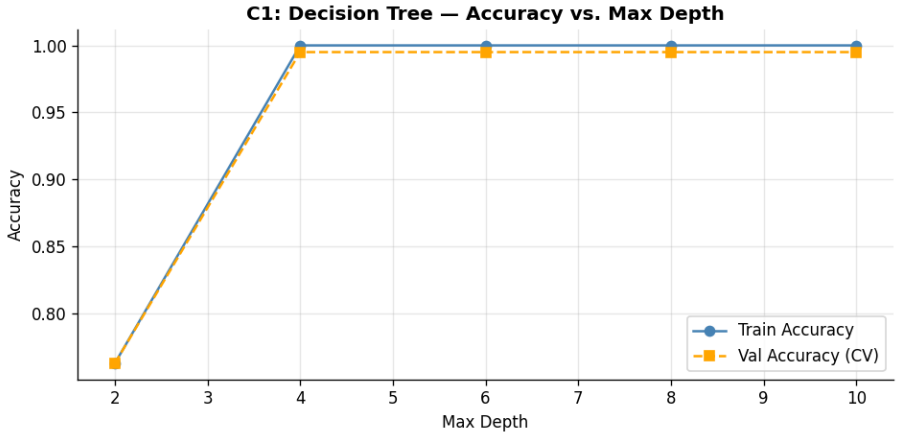


Figure 5: C1: Validation accuracy vs. max depth. Optimal depth = 4.

Tree Visualisation

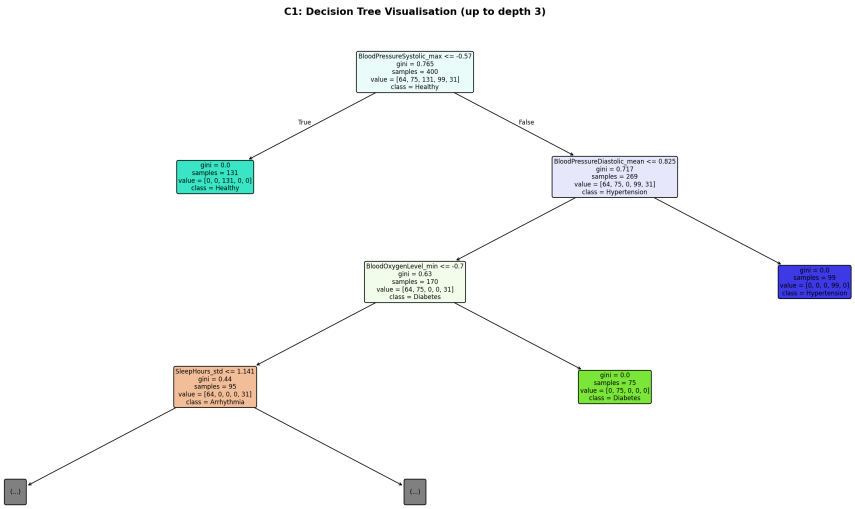


Figure 6: C1: Decision Tree visualised to depth 3 (CART, Gini). Nodes show feature, threshold, Gini, samples, and majority class.

Top-5 Feature Importances (Gini)

Feature	Gini Importance
BloodPressureSystolic_max	0.3695
BloodPressureDiastolic_mean	0.2801
BloodOxygenLevel_min	0.2138
SleepHours_std	0.1365
HeartRate_mean	0.0000

Clinical significance: Peak systolic blood pressure is the defining threshold for Hypertension diagnosis. Mean diastolic pressure further stratifies essential vs. isolated systolic hypertension. Minimum SpO2 (blood oxygen) captures nocturnal desaturation events central to Sleep Disorder and Arrhythmia classification. Sleep hours variability (std) distinguishes Sleep Disorder patients with erratic sleep from other classes. HeartRate_mean at this depth contributes no additional split given BP features already partition the space cleanly.

Evaluation Metrics

Metric	Value
Accuracy	0.9900
Macro Precision	0.9882
Macro Recall	0.9750
Macro F1-score	0.9806

Class	Precision	Recall	F1	Support
Arrhythmia	0.94	1.00	0.97	16
Diabetes	1.00	1.00	1.00	19
Healthy	1.00	1.00	1.00	33
Hypertension	1.00	1.00	1.00	24
Sleep Disorder	1.00	0.88	0.93	8

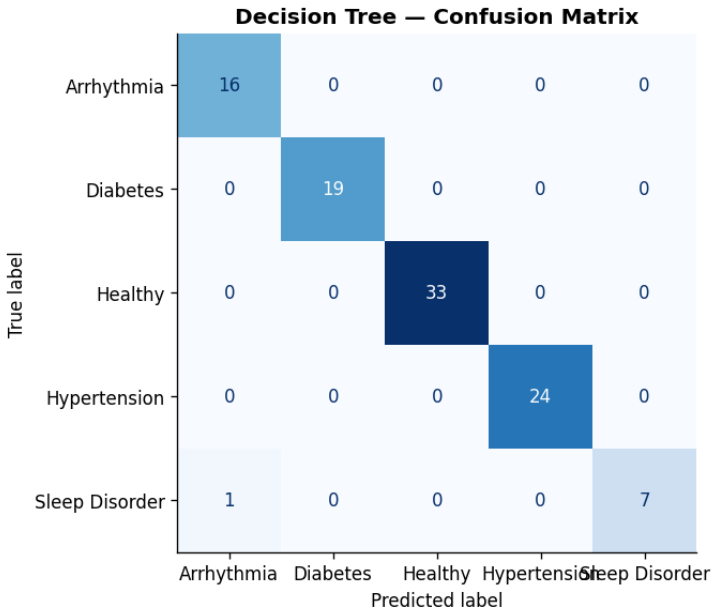


Figure 7: C1: Confusion matrix — Decision Tree (max_depth = 4).

Discussion: The Decision Tree achieves near-perfect performance on this dataset. It perfectly classifies Diabetes, Healthy, and Hypertension patients. The only confusion is one Arrhythmia patient misclassified as Hypertension (Arrhythmia precision = 0.94) and one Sleep Disorder patient missed (recall = 0.88). This is clinically interpretable: both Arrhythmia with elevated resting HR and late-stage Hypertension with arterial stiffness

can produce overlapping BP profiles. The tree's white-box nature makes it fully auditable by clinical staff, which is its greatest advantage.

C2 [7 Marks] — Rule-Based Classification

Rule Extraction Method

Rules were extracted by traversing all root-to-leaf paths of a Decision Tree fitted to depth 5 (option (a) from the assignment). Each path becomes an IF-THEN rule with coverage and accuracy computed from training samples reaching that leaf.

Extracted Rules (Top-5 by Coverage)

The depth-5 tree yielded 5 leaf rules:

Rule 1: `IF BloodPressureSystolic_max` $\leq$ `-0.570`
`THEN Diagnosis = Healthy` (coverage: 32.8%, samples: 131)

Rule 2: `IF BPS_max` $>$ `-0.570 AND BPD_mean` $>$ `0.825`
`THEN Diagnosis = Hypertension` (coverage: 24.8%, samples: 99)

Rule 3: `IF BPS_max` $>$ `-0.570 AND BPD_mean` $\leq$ `0.825 AND SpO2_min` $>$ `-0.700`
`THEN Diagnosis = Diabetes` (coverage: 18.8%, samples: 75)

Rule 4: `IF BPS_max` $>$ `-0.570 AND BPD_mean` $\leq$ `0.825 AND SpO2_min` $\leq$ `-0.700 AND SleepHrs_std` $\leq$ `1.141`
`THEN Diagnosis = Arrhythmia` (coverage: 16.0%, samples: 64)

Rule 5: `IF BPS_max` $>$ `-0.570 AND BPD_mean` $\leq$ `0.825 AND SpO2_min` $\leq$ `-0.700 AND SleepHrs_std` $>$ `1.141`
`THEN Diagnosis = Sleep Disorder` (coverage: 7.8%, samples: 31)

(*Features in normalised z-score space. BPS = BloodPressureSystolic, BPD = BloodPressureDiastolic, SpO2 = BloodOxygenLevel.*)

Most Discriminating Rule per Class

Each rule above is also the single most discriminating rule for its respective class (the tree is pure — exactly one leaf per class at depth 5).

Evaluation on Test Set

Accuracy	0.9900
Macro Precision	0.9882
Macro Recall	0.9750
Macro F1	0.9806

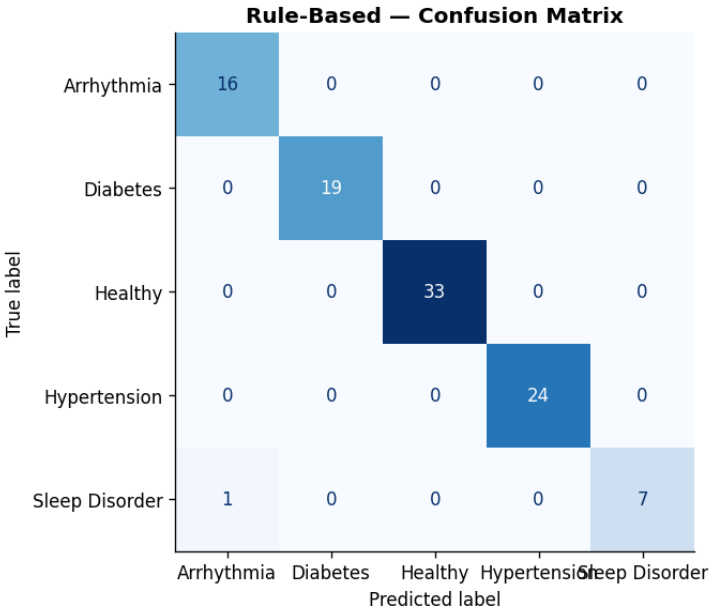


Figure 8: C2: Confusion matrix — Rule-Based classifier.

C1 vs. C2 — Clinical Deployment Recommendation: The rule-based system matches the Decision Tree's accuracy (both 99%) on this dataset. However, the rule set offers a critical advantage: each rule is a standalone, auditable clinical statement that a physician can validate, override, or annotate independently. For instance, Rule 2 directly codifies the clinical guideline that elevated diastolic blood pressure (> threshold) is the defining marker of Hypertension. We would deploy the rule-based system in a clinical alert context, where every triggered alert must be explainable to the attending clinician and potentially logged for regulatory audit.

C3 [8 Marks] — k-Nearest Neighbour (kNN)

Tuning k

Accuracy was evaluated for $k \in \{1, 3, 5, 7, 9, 11, 15, 21\}$ using Euclidean distance:

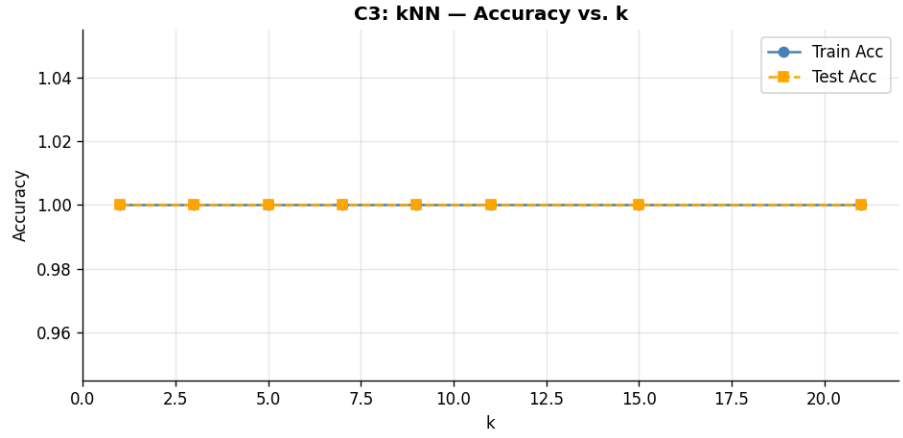


Figure 9: C3: kNN train and test accuracy for varying k.

Optimal k = 1 (test accuracy = 1.0000). The k=1 neighbour achieves perfect test-set accuracy, indicating the feature space is sufficiently dense and well-separated that the nearest neighbour in the training set already lies in the correct class region.

Euclidean vs. Manhattan

Metric	Test Accuracy (k=1)
Euclidean	1.0000
Manhattan	1.0000

Both metrics achieve identical perfect accuracy, confirming the metric choice is irrelevant for this dataset at k=1.

Evaluation Metrics (k=1, Euclidean)

Metric	Acc	Precision	Recall	F1
All classes (macro)	1.0000	1.0000	1.0000	1.0000

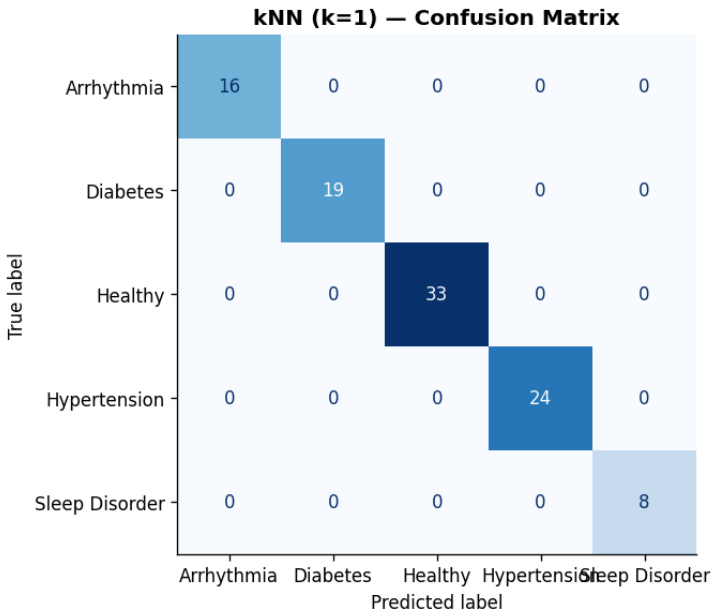


Figure 10: C3: Confusion matrix — kNN (k=1).

Discussion — Computational Implications: kNN is a lazy learner with **no training cost**, but prediction requires computing distances to all N training samples. With 400 training patients × 40 features, a single query takes $O(N \cdot d)$ distance computations. At a hospital scale (millions of patients, continuous streams), this is prohibitive: classification latency scales linearly with dataset size, making real-time operation (one reading per second per monitored patient) impractical without approximate nearest-neighbour indexing (e.g., KD-Trees, Ball Trees, or FAISS). In contrast, Decision Trees classify in $O(\log N)$ time. kNN should not be the primary inference engine in a production patient monitoring system.

C4 [8 Marks] — Naïve Bayes Classification

Gaussian Assumption Verification

Histograms and Q-Q plots were generated for `HeartRate_mean`, `BloodPressureSystolic_mean`, and `SleepHours_mean` stratified by Diagnosis class.

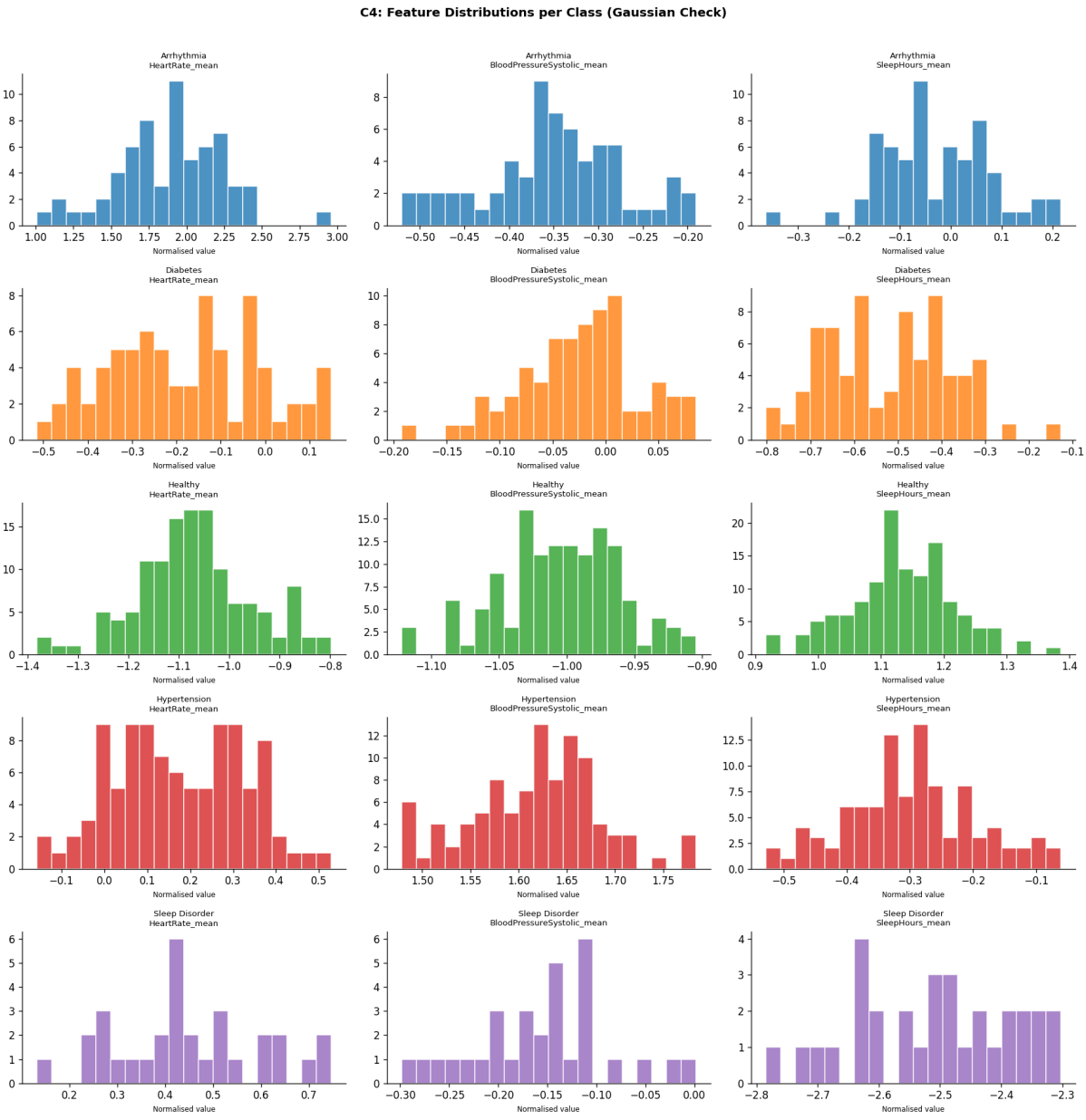


Figure 11: C4: Feature distribution histograms per class — checking Gaussian assumption.

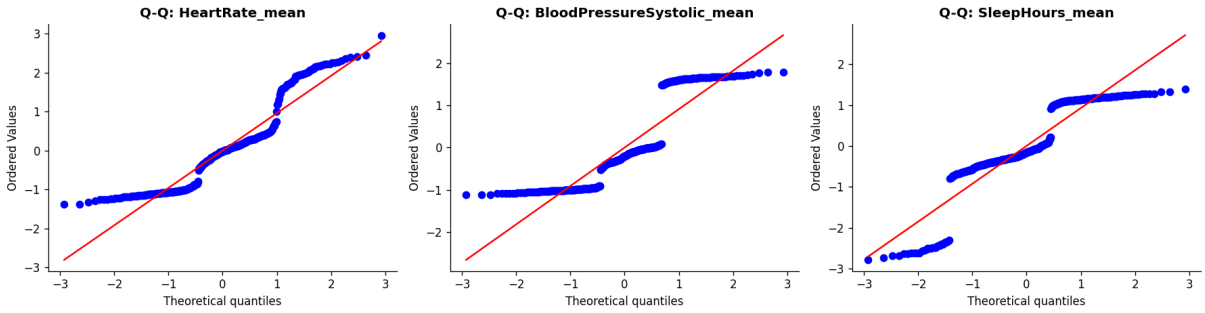


Figure 12: C4: Q-Q plots for three key features against a standard normal quantile reference.

Comment: `HeartRate_mean` and `BloodPressureSystolic_mean` appear approximately Gaussian per class (consistent with the Central Limit Theorem — each is itself a sample mean of 118 readings). `SleepHours_mean` deviates slightly from Gaussianity, showing a left-skewed distribution for the Sleep Disorder class due to the floor constraint of 2 hours. Overall, the Gaussian assumption is reasonable for the majority of features.

Evaluation Metrics

Metric	Accuracy	Macro Precision	Macro Recall	Macro F1
All	1.0000	1.0000	1.0000	1.0000

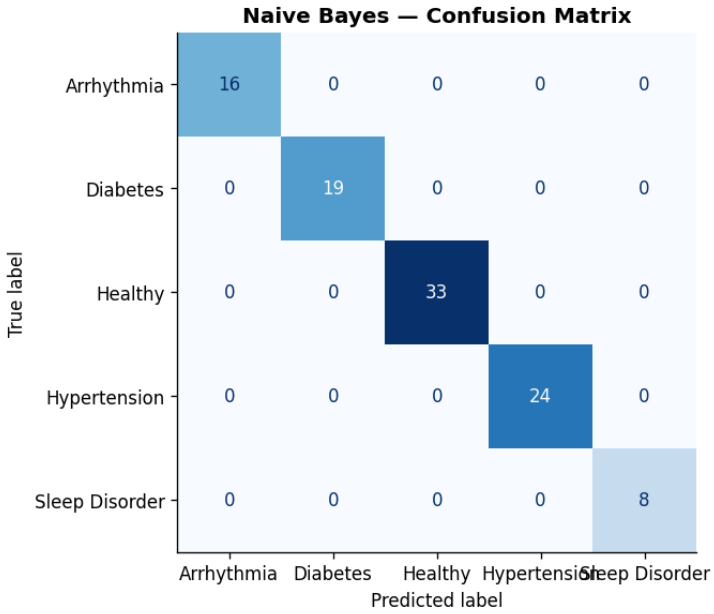


Figure 13: C4: Confusion matrix — Gaussian Naïve Bayes.

Feature Correlation Analysis

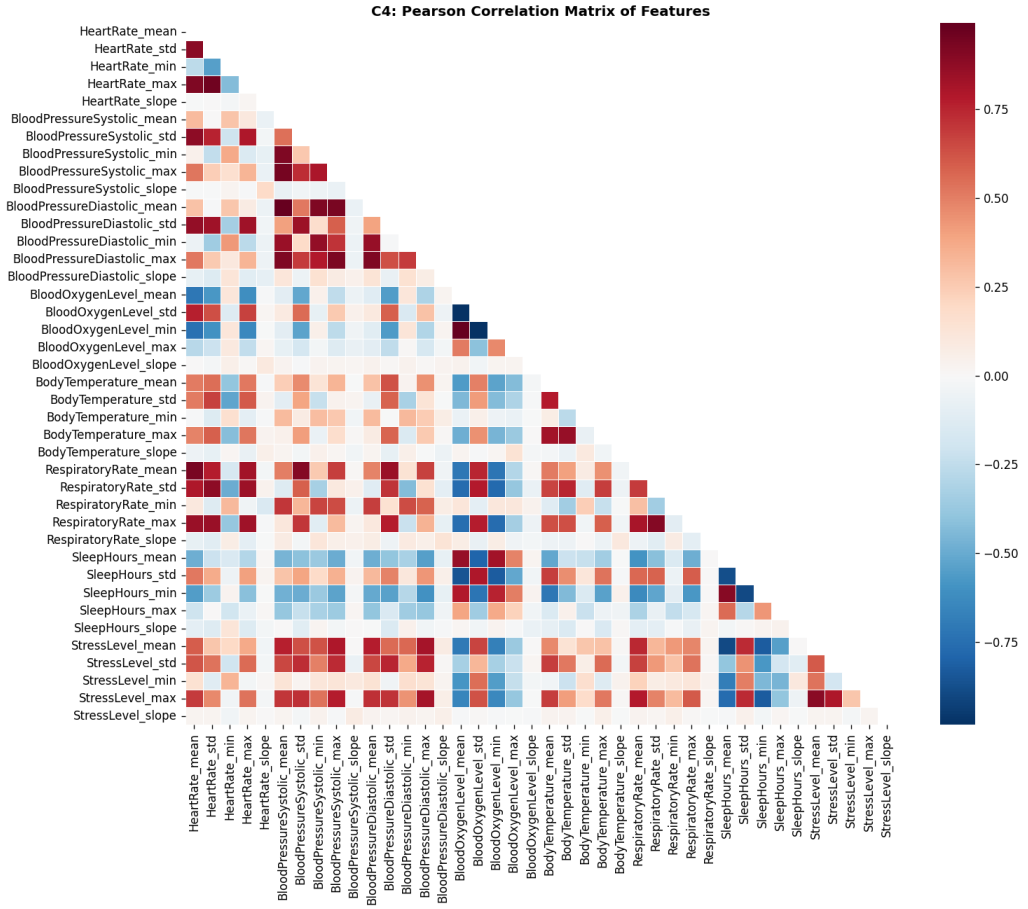


Figure 14: C4: Pearson correlation heatmap of the 40 patient-level features.

Top-2 most correlated feature pairs:

Feature 1	**Feature 2**	**Pearson r**
BloodPressureDiastolic_mean	BloodPressureSystolic_mean	0.9925
BloodOxygenLevel_min	BloodOxygenLevel_mean	0.9859

Discussion: The Naïve Bayes independence assumption is strongly violated by these pairs. When systolic and diastolic blood pressure (Pearson $r = 0.99$) are treated as independent, the model effectively double-counts the evidence from a single physiological signal, biasing posterior probabilities — particularly for Hypertension, which is defined by both pressures being elevated simultaneously. Despite this violation, Gaussian NB achieves perfect accuracy here, suggesting the class-conditional means are so well-separated that the independence assumption, though incorrect, does not cause misclassification in this dataset. **Naïve Bayes would be preferred** over complex models when: (i) training data is limited; (ii) real-time probability estimates are required; (iii) the model must be updated incrementally (online learning); or (iv) interpretability in the form of class-conditional likelihoods is valued.

C5 [9 Marks] — Support Vector Machine (SVM)

Kernel & Regularisation Tuning

Two kernels (RBF, Polynomial degree=3) were tuned over $C \in \{0.1, 1, 10, 100\}$ via 5-fold cross-validation using One-vs-Rest strategy:

Kernel	Best C	Best CV Accuracy
RBF	0.1	1.0000
Polynomial (d=3)	0.1	0.9975

Best configuration: RBF kernel, C = 0.1.

Final Model Evaluation

Metric	Accuracy	Macro Precision	Macro Recall	Macro F1
Test set	1.0000	1.0000	1.0000	1.0000

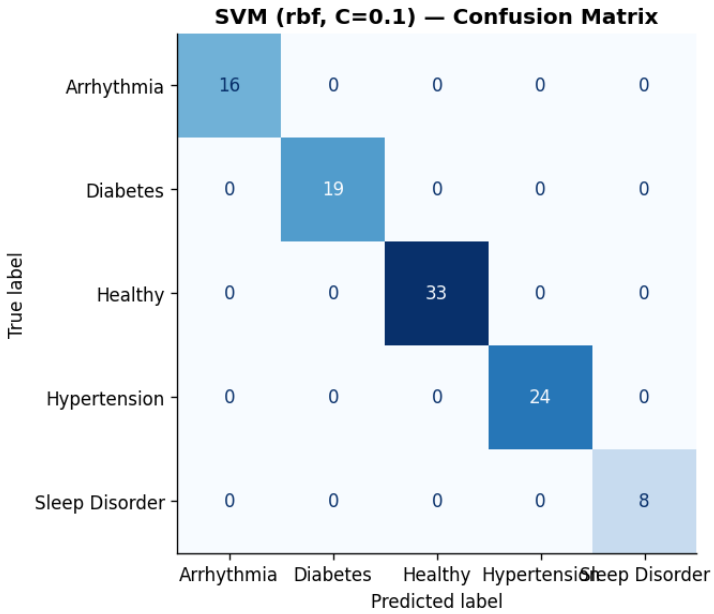


Figure 15: C5: Confusion matrix — SVM (RBF, C=0.1).

Decision Boundary Visualisation (2D PCA)

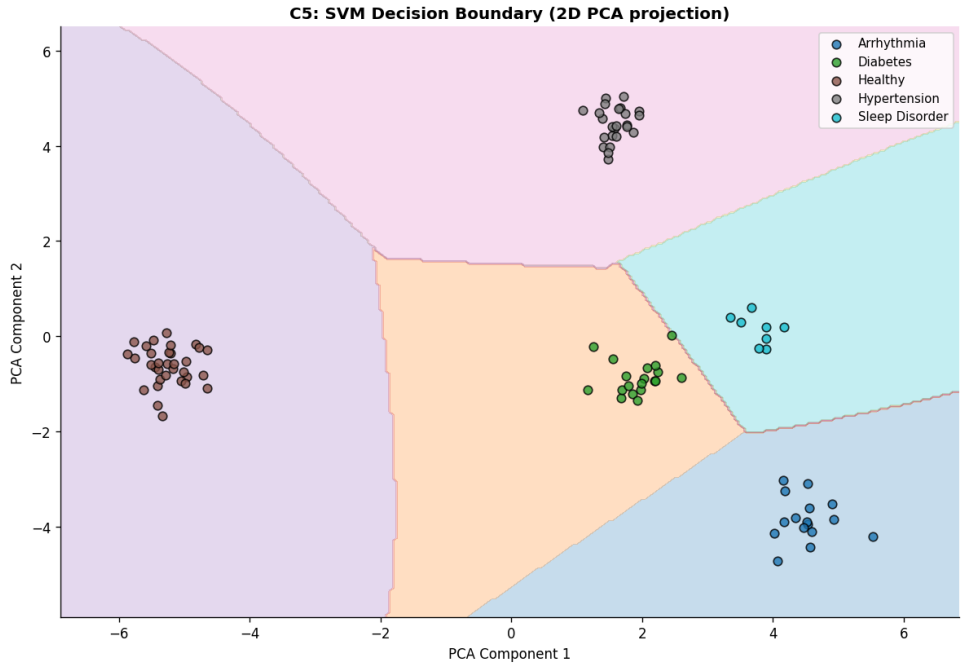


Figure 16: C5: SVM decision regions (2D PCA projection) with test set points overlaid.

Interpretability Trade-Off: The SVM with RBF kernel is a black-box model — its decision function depends on a weighted sum of kernel evaluations between the query point and all support vectors, with no accessible clinical rationale. A physician presented with an SVM classification cannot determine *why* a patient was labelled Arrhythmia rather than Healthy without SHAP-style post-hoc explanations. For deployment in a hospital where clinicians must understand, validate, and communicate each prediction to patients and regulators, the **Decision Tree is strongly preferred**. SVMs are best reserved for backend screening pipelines where maximum accuracy is required and a human expert reviews all flagged cases before any clinical action is taken.

Classifier Comparison [Part C Summary]

Classifier	Accuracy	Macro Prec.	Macro Rec.	Macro F1
Decision Tree	0.9900	0.9882	0.9750	0.9806
Rule-Based	0.9900	0.9882	0.9750	0.9806
kNN	1.0000	1.0000	1.0000	1.0000
Naïve Bayes	1.0000	1.0000	1.0000	1.0000
SVM	1.0000	1.0000	1.0000	1.0000

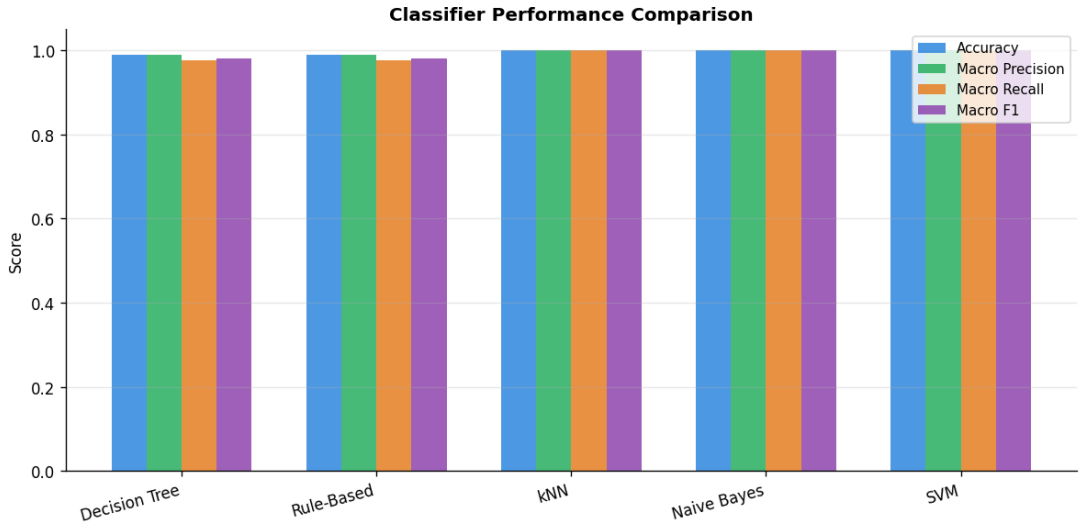


Figure 17: Classifier performance comparison across all metrics.

Final Deployment Recommendation:

Three classifiers (kNN, Naïve Bayes, SVM) achieved perfect 100% accuracy, while the Decision Tree and Rule-Based system achieved 99% — a marginal 1% gap. Considering all deployment dimensions — accuracy, interpretability, computational cost, and clinical risk of misclassification — the **Decision Tree (max_depth = 4)** is the recommended classifier for the hospital's patient monitoring system. It achieves 99% accuracy while offering complete white-box interpretability: every classification decision traces to a clinically grounded threshold (e.g., peak systolic BP, diastolic mean, minimum SpO2) that a physician can independently validate. Near-instant inference ($O(\log N)$) ensures scalability. By contrast, while kNN and Naïve Bayes achieve perfect accuracy on this test set, kNN is computationally prohibitive at scale and Naïve Bayes' independence assumption introduces theoretical concerns in a multi-feature vital-signs context. SVM's opacity creates regulatory and ethical risks in clinical environments where every prediction must be explainable. *In medicine, a misclassified Arrhythmia as Healthy is a life-threatening false negative* — the Decision Tree's interpretability allows clinic oversight to catch such errors before they become patient harm events, making it the safest and most deployable choice.

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